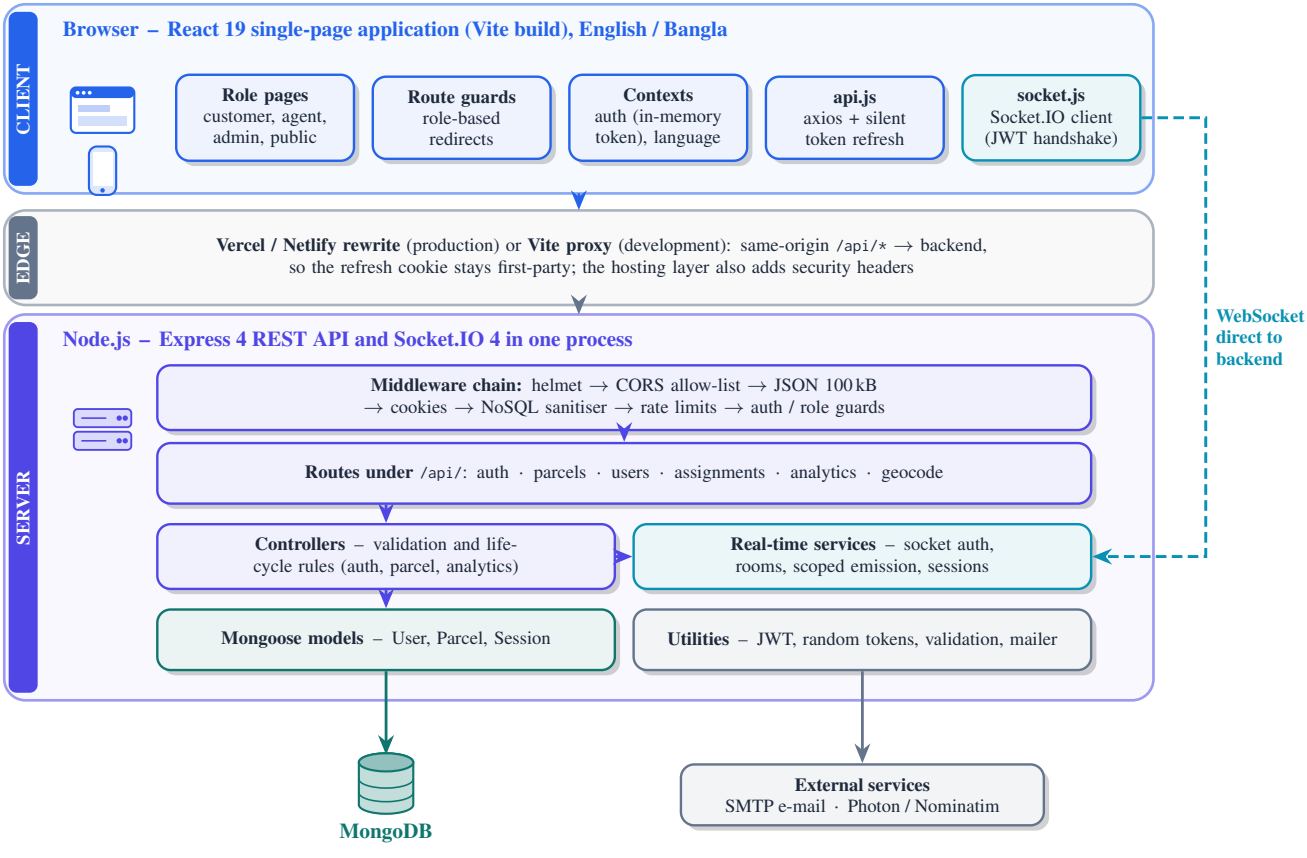
CLIENT
Browser – React 19 single-page application (Vite build), English / Bangla
Role pages
customer, agent, admin, public
Route guards
role-based redirects
Contexts
auth (in-memory token), language
api.js
axios + silent token refresh
socket.js
Socket.IO client (JWT handshake)
EDGE
Vercel / Netlify rewrite (production) or Vite proxy (development): same-origin /api/* → backend,
so the refresh cookie stays first-party; the hosting layer also adds security headers
WebSocket direct to backend
SERVER
Node.js – Express 4 REST API and Socket.IO 4 in one process
Middleware chain: helmet → CORS allow-list → JSON 100 kB
→ cookies → NoSQL sanitiser → rate limits → auth / role guards
Routes under /api/: auth · parcels · users · assignments · analytics · geocode
Controllers – validation and life-cycle rules (auth, parcel, analytics)
Real-time services – socket auth, rooms, scoped emission, sessions
Mongoose models – User, Parcel, Session
Utilities – JWT, random tokens, validation, mailer
MongoDB
External services
SMTP e-mail · Photon / Nominatim